MediCare

MAD-1

PROJECT REPORT

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Project Details

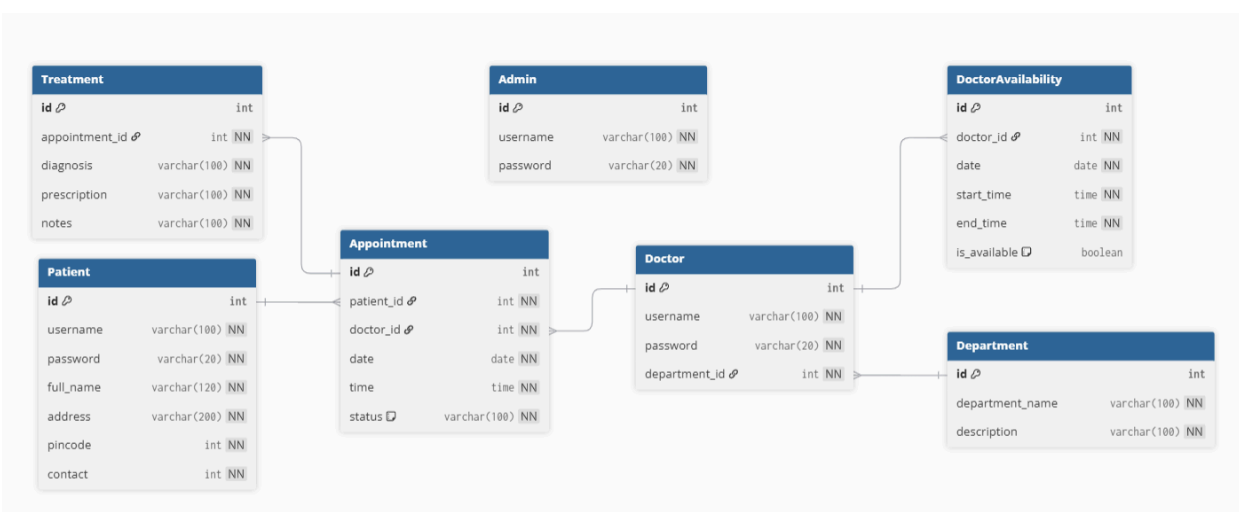
MediCare: Hospital Management System

MediCare is a comprehensive, multi-user hospital management application designed to facilitate efficient administration and easier appointment bookings. The system features three distinct roles: Admin, Doctor, and Patient. Admins can manage hospital departments, register doctors, and oversee all patient and appointment data via a central dashboard. Doctors can manage their availability schedules, view upcoming appointments, and record patient treatments (diagnoses and prescriptions). Patients can register, search for doctors by department, book appointments based on real-time availability, and access their medical history. It ensures seamless coordination between hospital staff and patients, making healthcare access organized and reliable.

Technologies Used

- **Flask:** Lightweight Python web framework to handle the backend API and routing.
- **Bootstrap :** CSS framework for building a responsive, mobile-friendly user interface.
- **Flask-SQLAlchemy:** ORM (Object Relational Mapper) for managing database interactions.
- **Jinja2:** Templating engine for rendering dynamic HTML pages.
- **SQLite:** Lightweight relational database for storing user data, appointments, and medical records.
- **HTML/CSS:** Used for the structure and styling of the application pages.

DB Schema



File Structure

```
Hospital-Management-App---V1/
├── instance/
│   └── site.db
├── static/
│   └── icu.avif
├── templates/
│   ├── admin_dashboard.html
│   ├── appointments.html
│   ├── doctor_dashboard.html
│   ├── doctors.html
│   ├── edit_doctor.html
│   ├── edit_patient.html
│   ├── index.html
│   ├── login.html
│   ├── manage_availability.html
│   ├── patient_dashboard.html
│   ├── patient_history.html
│   ├── patients.html
│   └── signup.html
├── app.py
├── config.py
├── models.py
├── routes.py
├── Project_report.pdf
└── README.md
```

API Resource / Routes

Authentication

- **Home**: /
- **Login**: /login
- **Signup**: /signup
- **Logout**: /logout

Admin Operations

- **Dashboard**: /admin_dashboard
- **Add Department**: /add_department
- **Add Doctor**: /add_doctor
- **View/Search Doctors**: /view_doctors
- **Edit/Delete Doctor**: /edit_doctor/<id>, /delete_doctor/<id>
- **View/Search Patients**: /view_patients
- **Edit/Delete Patient**: /edit_patient/<id>, /delete_patient/<id>
- **View All Appointments**: /view_appointments

Doctor Operations

- **Dashboard**: /doctor_dashboard
- **Manage Schedule**: /manage_availability
- **Complete Treatment**: /complete_appointment/<appt_id>
- **Cancel Appointment**: /doctor_cancel_appointment/<appt_id>
- **Patient History**: /patient_history/<patient_id>

Patient Operations

- **Dashboard**: /patient_dashboard
- **Book Appointment**: /book_appointment
- **Reschedule**: /reschedule_appointment/<appt_id>
- **Cancel**: /cancel_appointment/<appt_id>
- **Update Profile**: /update_patient_profile

Use of AI

I used a few AI tools to help me build this project. They were really useful for speeding up debugging and getting a hand with the Bootstrap layout, which made the whole process much smoother.

Demo Video link - [MediCare]